SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1516

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**

Question Count: 5

Code of Conduct

I _____VINIL KUMAR S (192511226)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

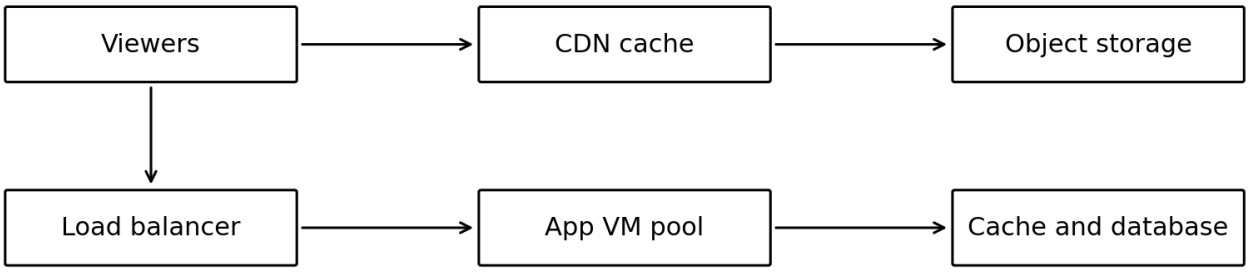
Scenario-Based Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

Question 1: Video Streaming Cloud Architecture

Scenario and major risks

The platform must serve millions of concurrent viewers with minimal buffering. Main risks are traffic spikes, overloaded origin servers, network delay, insufficient bandwidth, cache misses and service outages. Unequal resource use by virtual machines can also reduce streaming quality.



Architecture and caching

Store video files in durable object storage and transcode them into multiple resolutions using queued worker jobs. A content delivery network (CDN) caches video segments near viewers. Adaptive bitrate streaming adjusts video quality to the available bandwidth. Cache popular segments for longer periods and use versioned file names to avoid serving outdated content. Keep account and payment responses private and outside shared caches.

Load balancing and auto scaling

Place stateless application instances on virtual machines or containers across multiple availability zones. A load balancer distributes requests only to healthy instances. Scale application instances using request rate, CPU use and response latency; scale transcoding workers using queue length. Keep minimum spare capacity and pre-scale for scheduled live events because starting new instances takes time. External session storage allows requests to reach any healthy instance.

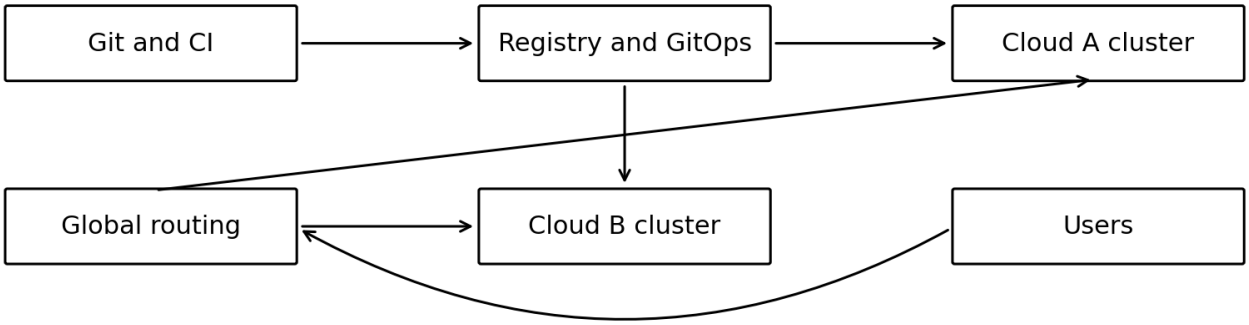
Cost optimization and justification

CDN caching reduces repeated origin reads and origin bandwidth. Right-size instances and set scaling limits with budget alerts. Use committed capacity for predictable demand and interruptible instances only for retryable transcoding jobs. Apply storage lifecycle policies to older originals and monitor delivery charges. CDN capacity handles most video traffic, while horizontally scaled compute handles authentication and metadata. Load tests should measure startup delay, buffering ratio, cache hit rate and application latency to verify the design.

Question 2 : Multi Cloud Container Deployment

Scenario and major risks

The startup needs consistent container deployments across cloud providers while protecting financial information. Main risks include configuration drift, incompatible services, exposed secrets, container vulnerabilities, cross-cloud latency and inconsistent transaction data. Containers share a host kernel, so container isolation alone is insufficient for sensitive workloads.



Orchestration and deployment

Deploy a managed Kubernetes cluster in each cloud instead of stretching one cluster across unreliable wide-area links. Package each application as a versioned container image and deploy the same immutable image digest using Helm and GitOps. Store reviewed manifests in Git and use automated tests, image scanning and controlled rolling updates. Use infrastructure as code for networks and cluster settings. Apply resource requests and limits, network policies, non-root execution and patched worker VMs to improve isolation and predictable performance.

Service discovery and scaling

Within each cluster, Kubernetes Services and internal DNS provide stable names while pods change. Readiness checks keep unready pods out of traffic, and a load balancer or ingress routes external requests. Horizontal Pod Autoscaling adds replicas based on CPU or workload metrics; a node autoscaler adds worker capacity when pods cannot be scheduled. Use global DNS or traffic management for cross-cloud routing. Cross-cloud service discovery requires explicit configuration; internal cluster DNS does not automatically span providers.

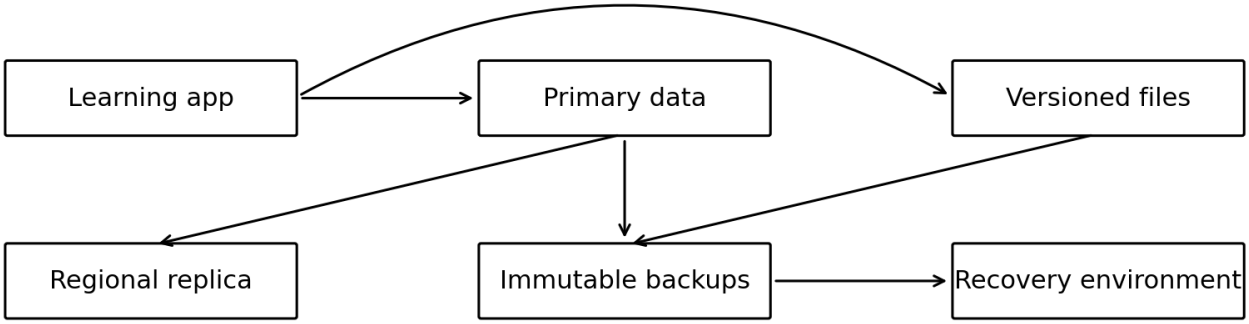
Benefits risks and justification

Multi-cloud reduces dependence on one provider and supports regional availability and service choice. However, it adds operational effort, transfer costs, policy differences and failure modes. Keep the transactional database under a clearly defined primary and use replicated recovery copies with measured lag; independent writable copies can create financial inconsistencies. Enforce centralized identity, secrets management, encryption and audit logs. Separate clusters limit failures to one provider while common deployment definitions reduce drift. Test provider failover and payment retry idempotency before production.

Question 3: Backup and Disaster Recovery for E Learning

Scenario and major risks

Accidental deletion can remove course files, student submissions or database records. Live replication alone cannot solve this because it can copy the deletion to another location. Other risks include ransomware, compromised administrator credentials, corrupted backups and recovery procedures that have never been tested.



Backup replication and versioning

Enable object versioning so earlier file versions remain recoverable, and apply suitable retention to deleted versions. Schedule cloud-native database backups and transaction-log capture for point-in-time recovery. Keep encrypted immutable backups in a separate account and replicate recovery copies to another region. Restrict backup deletion through separate roles and protect encryption keys. Replicate application data for infrastructure failure recovery, while retaining historical backups for logical errors. Store infrastructure definitions and application configurations with the recovery plan.

RTO and RPO targets

RTO is the maximum acceptable time to restore service; RPO is the maximum acceptable amount of data loss measured in time. Proposed targets are an RTO of 60 minutes and an RPO of 15 minutes for critical student records. Capture transaction logs at intervals shorter than 15 minutes and monitor their arrival in the recovery region. Maintain warm standby capacity and automate restoration to support the RTO. Actual recovery time includes detection, provisioning, restoration, validation and traffic switching. These targets must be demonstrated through timed drills; a daily backup alone cannot achieve a 15-minute RPO.

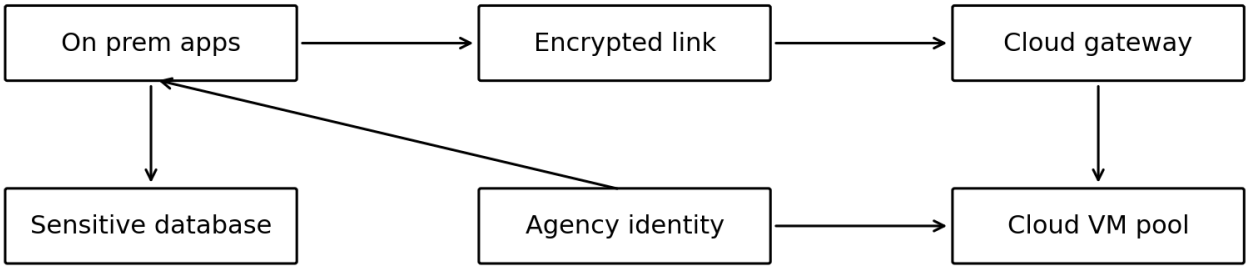
Business continuity and justification

First detect the incident and stop harmful writes. Identify the last clean recovery point and restore into an isolated environment. Validate record counts, file integrity and application access, then switch traffic to the recovered service. Prioritize login, course access and submissions; communicate temporary restrictions and reconcile later changes before failback. Conduct scheduled restore tests and record achieved RTO and RPO. Versioning handles accidental deletion, immutable backups resist malicious changes, and regional recovery protects against wider outages.

Question 4 : Secure Hybrid Cloud Architecture

Scenario and major risks

The agency must retain regulated applications and sensitive records on premises while using public cloud resources for permitted workloads. Main risks include unauthorized data movement, interception, inconsistent access policies, network outages, misconfiguration and exposure through shared cloud infrastructure.



Hybrid architecture and communication

Keep sensitive databases and restricted applications in an on-premises private cloud using isolated VMs and network segments. Host public websites and approved non-sensitive services in a public cloud virtual network. Connect the environments through redundant site-to-site IPsec VPNs or an encrypted dedicated connection. Route approved requests through an API gateway and firewall; avoid direct public access to internal databases. Apply data classification and filtering so only authorized fields cross the boundary. Public cloud deployment must not bypass the requirement to retain restricted data locally.

Identity management and encryption

Federate the agency identity provider with cloud identity services for single sign-on. Require multifactor authentication for staff, role-based access and least privilege. Give applications short-lived workload identities instead of shared passwords. Use TLS for application traffic and encryption at rest for databases, VM disks and object storage. Protect keys with a key management service or hardware security module, separate key administrators from data users and rotate keys according to policy. A dedicated network connection is not automatically encrypted, so encryption must be explicitly configured.

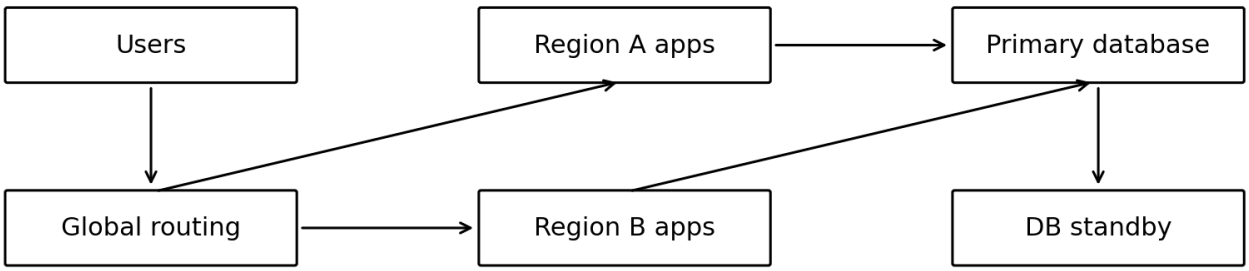
Management challenges and justification

Hybrid environments require coordinated patching, monitoring, backup and policy enforcement across two infrastructures. Network latency can slow applications that repeatedly query on-premises databases, while link failure can interrupt cloud services. Keep tightly coupled sensitive components together, cache only approved non-sensitive data and provide redundant links. Use infrastructure as code, configuration audits and centralized security logs to reduce drift. Scan hypervisors and guest systems and set VM resource limits to reduce contention. This design uses cloud elasticity for public workloads while preserving the location and access controls required for sensitive information.

Question 5 : Highly Available Multi Region SaaS

Scenario and major risks

The SaaS application must remain available when an entire region fails. Main risks are regional outages, shared service dependencies, stale replicas, failed traffic switching, duplicate writes and insufficient recovery capacity. Availability zones alone cannot protect against the loss of their entire region.



Multi region deployment

Run stateless application instances across multiple zones in two regions. Use global health-aware routing and a regional load balancer in each location. Maintain enough capacity or reserved headroom in the second region to absorb the failed region's load. Replicate object storage, configuration and required secrets; avoid dependence on an identity or storage service available only in the failed region. VMs or containers provide consistent deployment units, while replicas across independent regions reduce the impact of infrastructure failure.

Data replication and failover

Use a single writable database primary with a cross-region standby and keep writes directed to the primary. Asynchronous replication reduces normal write latency but may lose recent unreplicated transactions during a disaster. Define an acceptable RPO and monitor replication lag. If zero committed-data loss is required, evaluate synchronous cross-region replication and its latency and availability tradeoffs. During failover, fence the old primary, confirm replica readiness, promote the standby and switch application connections. Fencing prevents split-brain writes. Use idempotent requests to prevent duplicate processing during retries. Reconcile data before controlled failback.

Monitoring alerts and justification

Collect application latency, error rate, request volume, VM or container saturation, database health and replication lag from both regions. Run external synthetic checks of login and a core user journey. Alert the on-call team when user-impact thresholds or recovery limits are breached; route alerts through infrastructure outside the failed region. Test regional evacuation and record achieved RTO and RPO. Global routing removes unhealthy endpoints, but cached DNS can delay redirection, so account for TTL and client behaviour. This approach provides regional resilience while making the consistency and recovery tradeoffs explicit.